

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1637

Joliet 100 kV
November 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1637. The Interconnection Customer proposes a 221.7 MW solar photovoltaic facility interconnecting at the Joliet 100 kV substation in the Illinois area. The request was received on December 27, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 405 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,962,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Joliet-Red Bank 100 KV is overloaded to 135.5% of its 125 MVA emergency rating for the loss of the Lincoln 230 kV/1 kV XFMR. The project contributes 26.2% DFAX (58.0 MW) to this overload.

The E J Stoneman Sta-Moorhead 100 KV #2 is overloaded to 118.6% of its 144 MVA emergency rating for the loss of the E J Stoneman Station-Moorhead 100 KV. The project contributes 10.7% DFAX (23.6 MW) to this overload.

The Wilton-Lincoln 100 KV #2 is overloaded to 117.3% of its 186 MVA emergency rating for the loss of the Wilton-Lincoln 100 KV #1. The project contributes 10.4% DFAX (23.0 MW) to this overload.

The Lincoln-Wilton 100 KV #1 is overloaded to 113.0% of its 193 MVA emergency rating for the loss of the Wilton-Lincoln 100 KV. The project contributes 10.4% DFAX (23.0 MW) to this overload.

For the loss of E J Stoneman Sta-Air Products 100 KV Line 2, the E J Stoneman Station-Air Products 100 KV Circuit 1 loads to 112.3% of its 161 MVA rating. Project DFAX: 10.4%.

The Air Products-Wilton 100 KV Circuit 2 is overloaded to 112.1% of its 178 MVA emergency rating for the loss of the Wilton-Air Products 100 KV Ckt 1. The project contributes 10.3% DFAX (22.8 MW) to this overload.

For the loss of E J Stoneman Sta-Moorhead 100 KV Line 2, the E J Stoneman Sta-Moorhead 100 KV loads to 110.9% of its 154 MVA rating. Project DFAX: 10.7%.

For the loss of E J Stoneman Station-Air Products 100 KV #1, the Air Products-E J Stoneman Station Substation 100 KV Line 2 loads to 108.5% of its 167 MVA rating. Project DFAX: 10.4%.

The Wilton-Air Products 100 KV Line 1 is overloaded to 106.5% of its 187 MVA emergency rating for the loss of the Air Products Substation-Wilton 100 KV #2. The project contributes 10.3% DFAX (22.8 MW) to this overload.

For the loss of Moorhead-Bringhurst 100 KV Ckt 2, the Bringhurst-Moorhead 100 KV Ckt 1 loads to 103.6% of its 140 MVA rating. Project DFAX: 10.4%.

Bringhurst-Moorhead 100 KV Circuit 2: 103.6% loading (140 MVA) following the outage of the Moorhead-Bringhurst 100 KV Cir. 1; DFAX = 10.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Joliet-Red Bank 100 KV	Rebuild line with higher-capacity conductor	\$2,841,000	89% (\$2,516,000)
E J Stoneman Sta-Moorhead 100 KV #2	Rebuild line with higher-capacity conductor	\$1,981,000	90% (\$1,774,000)
Wilton-Lincoln 100 KV #2	Rebuild line with higher-capacity conductor	\$2,600,000	100% (\$2,600,000)
Wilton-Lincoln 100 KV L	Rebuild line with higher-capacity conductor	\$2,904,000	100% (\$2,904,000)
E J Stoneman Station-Air Products 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$3,914,000	100% (\$3,914,000)
Wilton Substation-Air Products 100 KV Circuit 2	Rebuild line with higher-capacity conductor	\$4,587,000	100% (\$4,587,000)
E J Stoneman Sta-Moorhead 100 KV	Rebuild line with higher-capacity conductor	\$1,820,000	90% (\$1,630,000)
E J Stoneman Sta-Air Products 100 KV Cir. 2	Reconductor line	\$929,000	100% (\$929,000)
Wilton-Air Products 100 KV Cir. 1	Reconductor line	\$1,326,000	100% (\$1,326,000)
Bringinghurst-Moorhead 100 KV Cir. 1	Reconductor line	\$852,000	100% (\$852,000)
Bringinghurst Substation-Moorhead 100 KV Ckt 2	Reconductor line	\$930,000	100% (\$930,000)
	Total allocated to project		\$23,962,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.